UNSUPERVISED DEPTH PREDICTION FROM MONOCULAR IMAGES

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Outline

- Problem Definition.
- Previous work.
- What to improve?
- Results.
- Results analysis.
- Future work.

Problem Definition

Given: Sequence of monocular images captured from a camera mounted on a car.
Required: Predict the depth of each pixel in the images.

Figure: A sequence of monocular images.

Previous Work

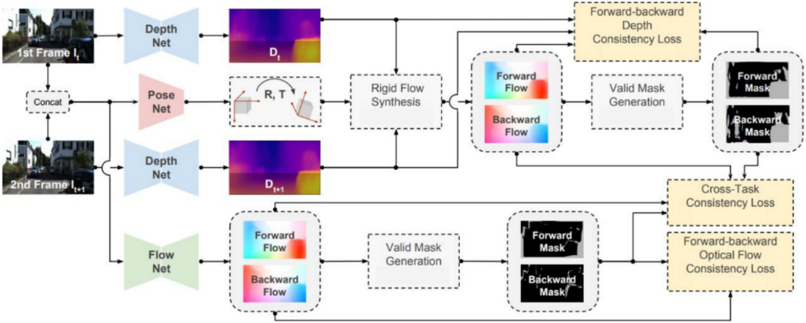
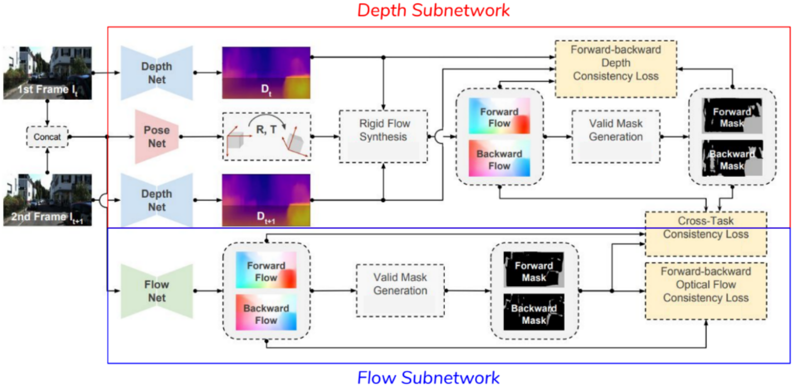


Figure: Current state of the art unsupervised framework for depth estimation.[9]

Depth and Flow Pipelines

The framework is divided into two subnetworks; Depth Subnetwork and the Flow subnetwork



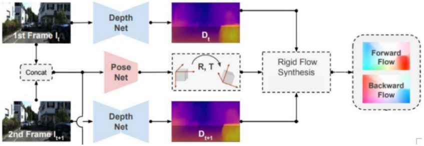
Rigid Flow Synthesis

Using dense pixel correspondence:

$$P_{t+1} = KT^{t \to t+1} D_{(}p_t) K^{-1} p_t \tag{1}$$

Obtain the rigid flow:

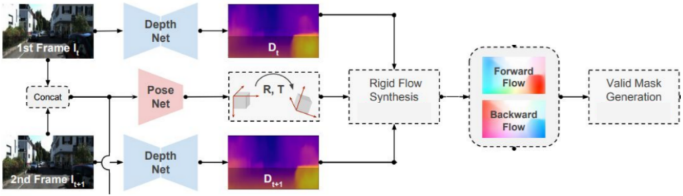
$$F_{rigid}(p_t) = p_{t+1} - p_t \tag{2}$$



Note: There are two flows obtained; one in each direction(from t to t+1 and vice versa).

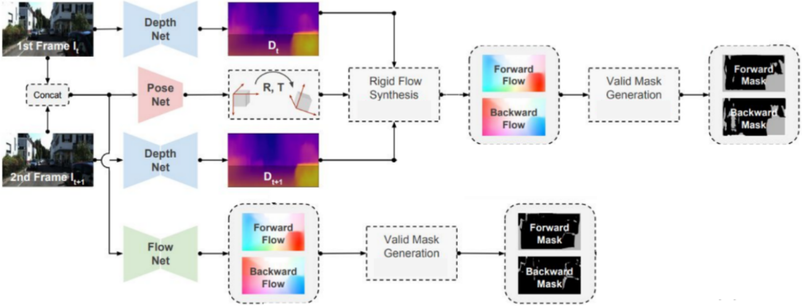
Valid Mask Generation

- Generating the flow forward and backward should arrive at the same position. Mask those pixels that violate this constraint.
- Ideally, the valid masks should contain all the pixels that resemble a dynamic object. Due to the movement induced from frame 1 to frame 2, all the pixels on the border should also be part of the valid masks accordingly.



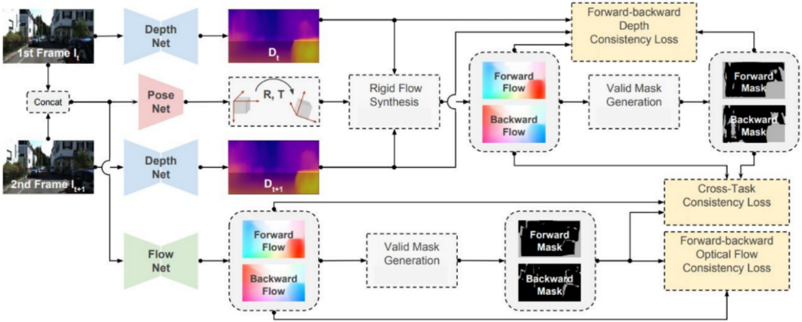
Flow subnetwork

Forward and backward flow as well as the valid mask are also generated from the flow pipeline



DF-Net Framework

There are five supervisory signals that are implemented; three of which can be seen below.



Total Loss, L

$$L = L_{photometric} + \lambda_s L_{smooth} + \lambda_f L_{forward-backward} + \lambda_c L_{cross} \quad (3)$$

Losses-Photometric and smoothness

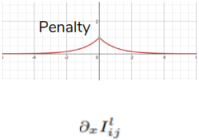
$$L = L_{\text{photometric}} + \lambda_s L_{\text{smooth}}$$

Given I_t, I_{t+1} and $F_{t\text{->}t+1}$, use inverse warping to generate $\bar{I}_t$

$$L_{\text{photometric}} = \sum_{p \in V} \rho\big(I_t(p), \bar{I}_t(p)\big).$$

V excludes the occluded and the dis-occluded regions

ρ(.) function is the ternary loss- Proved to handle better more complex illumination changes



$$C_{ds}^l = \frac{1}{N} \sum_{i,j} \left|\partial_x d_{ij}^l\right| e^{-\left\|\partial_x I_{ij}^l\right\|} + \left|\partial_y d_{ij}^l\right| e^{-\left\|\partial_y I_{ij}^l\right\|}.$$

Add loss whenever a change is depth exists but no gradient in image detected

Losses-Forward Backward

$$L_{\text{forward-backward, flow}} = \sum_{p \in V_{\text{flow}}} ||F_{t\to t+1}(p) + F_{t+1\to t}(p + F_{t\to t+1}(p))||_1$$

All valid points from the flow subnetwork

$$L_{\text{forward-backward, depth}} = \sum_{p \in V_{\text{depth}}} ||D_t(p) - \bar{D}_t(p)||_1$$

All valid points from the depth subnetwork

Given $F_{t\text{->}t+1}$, $F_{t+1\text{->}t}$ and D_{t+1}, use inverse warping to generate $\check{D}_t$

Losses-Consistency Loss

$$L_{\mathrm{cross}} = \sum_{p \in V_{\mathrm{depth}} \cap V_{\mathrm{flow}}} ||F_{\mathrm{rigid}}(p) - F_{\mathrm{flow}}(p)||_1$$

Union of valid points from the flow and depth networks

What to improve?

- Mask consistency: Design a loss that forces both forward and backward masks from the Depth subnetwork and the Flow subnetwork to match.
- Mask Edge Aware Loss: Apply a loss on the borders of each cluster in the mask when no image gradient is observed.

Mask Consistency Loss

- ideally we want:

$$V_{t \to t+1}^{depth} = V_{t \to t+1}^{flow} \tag{4}$$

$$V_{t+1 \to t}^{depth} = V_{t+1 \to t}^{flow} \tag{5}$$

- Unfortunately, the mask generation process is non-differentiable. Therefore, we opt to enforce the mask consistency through the flow. i.e:

$$L_{MC} = \sum_{p \in V_{depth} \neq V_{flow}} (F_{t \to t+1}^{depth} - F_{t+1 \to t}^{depth}) - (F_{t \to t+1}^{flow} - F_{t+1 \to t}^{flow}) \tag{6}$$

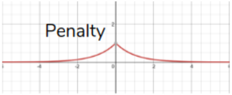
Mask Edge Aware Loss

- The mask is a result of the change in flow.
- Ideally, this flow should represent borders and dynamic objects.
- Remove the delta flow at the borders (still an ugly hack).

Figure: Mask removal Example

Mask Edge Aware Loss

Loss is added whenever there exits delta flow with no or very low change in image gradient.



$\partial_x I_{ij}^l$

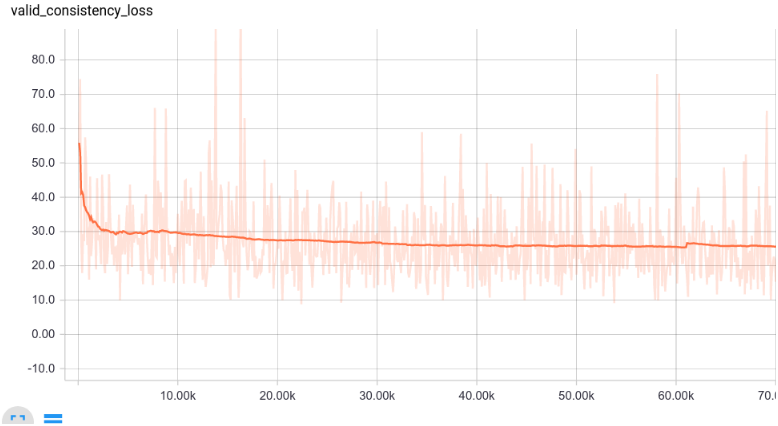
$$L_{ME} = \frac{1}{N} \sum_{i,j} |\partial_x \Delta F_{i,j}| e^{-||\partial_x I_{i,j}||} + |\partial_y \Delta F_{i,j}| e^{-||\partial_y I_{i,j}||}$$

$$\Delta F_{i,j} = F_{t \to t+1}^{depth} - F_{t+1 \to t}^{depth}$$

$$\Delta F_{i,j} = F_{t \to t+1}^{flow} - F_{t+1 \to t}^{flow}$$

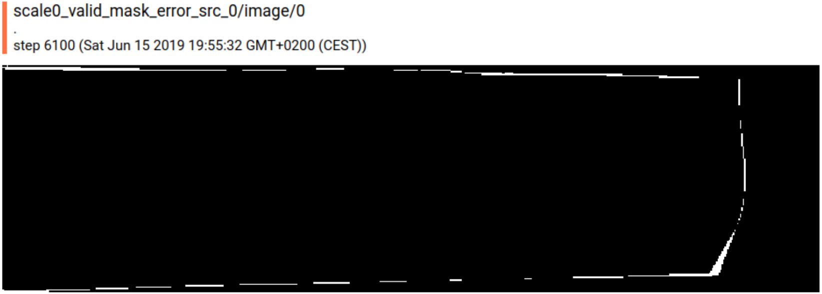
Results: Valid Consistency Loss

The loss of the Valid Consistency Loss decreases with every step.



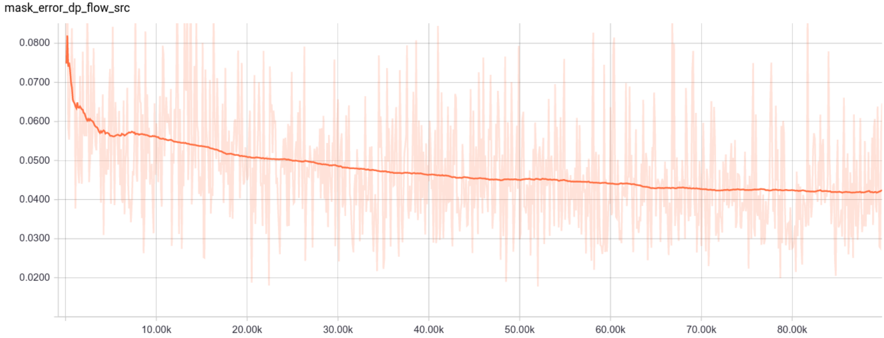
Snippets from masks

The difference between the masks computed from the Depth and Flow subnetwork for one image is computed. The pixel positions where the masks from both subnetworks do not match is resembled in white.



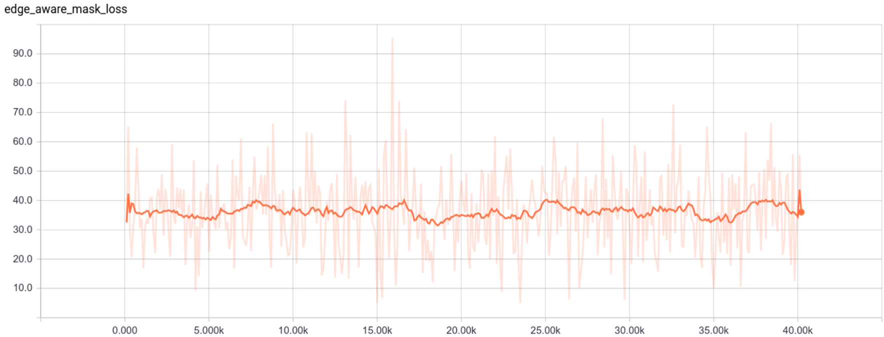
Results: Difference between masks

The percentage of mismatched pixels between masks computed from the Depth and Flow subnetwork is computed at each step. The percentage decreases from 8% to almost 4%.



Results: Edge aware mask loss

The EAM loss is displayed. We hypothesize that the reason the loss does not decay is due to the fact that the depth output in the beginning is poor. Therefore, the EAM does not converge in the desired manner. Schedule learning could help.



Results: output samples

The first row of images are the input images for our framework during test time whereas the second row is the output predicted depth.

Results Analysis: Error Metrics Explained

The following table formulates each of the error and accuracy metrics used to compare with state of the art results. x and x^* is the ground truth and estimated results. $x \in \{d,f\}$.

Abs Rel: $\frac{1}{\lvert D \rvert}\sum_{d' \in D} \lvert d^* - d' \rvert / d^*$	Sq Rel: $\frac{1}{\lvert D \rvert}\sum_{d' \in D} \lvert\lvert d^* - d' \rvert\rvert^2 / d^*$
RMSE: $\sqrt{\frac{1}{\lvert D \rvert}\sum_{d' \in D} \lvert\lvert d^* - d' \rvert\rvert^2}$	RMSE log: $\sqrt{\frac{1}{\lvert D \rvert}\sum_{d' \in D} \lvert\lvert \log d^* - \log d' \rvert\rvert^2}$
δ_t: % of $d \in D$ $max(\frac{d^*}{d}, \frac{d}{d^*}) < t$	
EPE: $\frac{1}{\lvert F \rvert}\sqrt{\sum_{f' \in F} \lVert f^* - f' \rVert^2}$	F1: err $>$ 3px and err $> \lvert f^* \rvert \times 5\%$

Results Analysis: Depth benchmark

Table: A comparison of the depth error metrics between our algorithm and the current state of the art is presented. Best results are highlighted in red. K refers to the KITTI dataset. The test set is preformed on kittiraw dataset.

Method	Dataset	Abs Rel	Sq Rel	RMSE	log RMSE
Zhou et al.[10]	K	0.208	1.768	6.856	0.283
Yang et al. [7]	K	0.182	1.481	6.501	0.267
Mahjourian et al. [3]	K	0.163	1.240	6.220	0.250
Yang et al. [6]	K	0.162	1.352	6.276	0.252
Yin et al. [8]	K	0.155	1.296	5.857	0.233
Godard et al. [1]	K	0.154	1.218	5.699	0.231
DF-Net [9]	K	0.150	1.124	5.507	0.223
Ours	K	0.146	1.158	5.353	0.214

Results Analysis: Depth benchmark

Table: A comparison of the depth accuracy metrics between our algorithm and the current state of the art is presented. Best results are highlighted in red. K refers to the KITTI dataset. The test set is preformed on kittiraw dataset.

Method	**Dataset**	$\delta < 1.25$	$\delta < 1.25^2$	$\delta < 1.25^3$
Zhou et al.[10]	K	0.678	0.885	0.957
Yang et al.[7]	K	0.725	0.906	0.963
Mahjourian et al. [3]	K	0.762	0.916	0.968
Yang et al.[6]	K	-	-	-
Yin et al.[8]	K	0.793	0.931	0.973
Godard et al. [1]	K	0.798	0.932	0.973
DF-Net [9]	K	0.806	0.933	0.973
Ours	K	0.816	0.942	0.978

Results Analysis: Flow benchmark

Table: A comparison of the flow error metrics between our algorithm and the current state of the art is presented. Best results are highlighted in red. K, C and SYN refer to KITTI, Cityscape and Synthia datasets. The test sets are kitti flow 2012(K12) and K15. Although other approaches are trained on other datasets, Ours is still able to perform better on the EPE metrics.

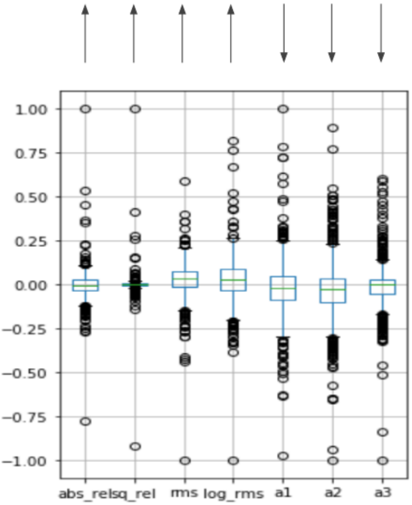
Method	Dataset	K12 EPE	K15 EPE	K15 F1
UnsupFlownet [2]	C + K	11.3	-	-
DSTFlow [5]	C	16.98	24.30	52.00
DSTFlow [5]	K	10.43	16.79	36.00
Yin et al. [8]	K	-	10.81	-
UnFlowC [4]	SYN + K	3.78	8.80	28.94
DF-Net [9]	SYN + K	3.54	8.98	26.01
Ours	K	2.50	8.75	27.62

Results Analysis: DF-Net Vs Ours

An in depth per sample comparison was performed to get a better insight on the pros and cons of the added losses.

- Each sample on the kittiraw depth set can be evaluated based on the aforementioned metrics.
- Subtracting the metrics of DF-Net's output from Ours followed normalizing the results and drawing a box plot is shown in the following slide.
- The arrows on top of each box plot resemble the direction of fitness. That is, an arrow pointing upward means that the higher the median of the box plot, the better our results are.
- A sample of the outliers on both ends of the box plot are withdrawn and displayed.

Results Analysis: DF-Net Vs Ours

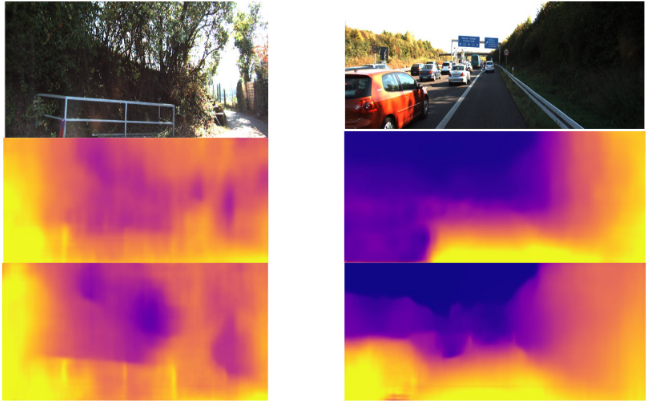


Depth Results Analysis: Better

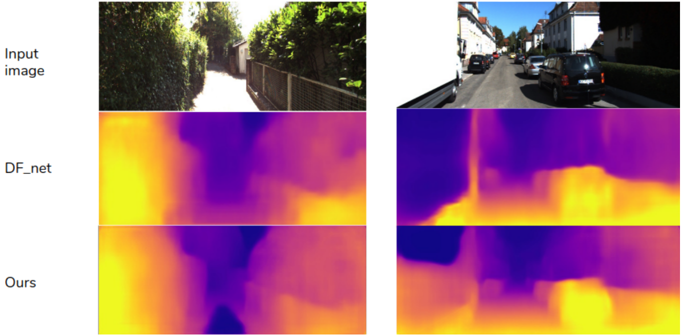
Input image

DF_net

Ours



Depth Results Analysis: Worse



Depth Results Analysis

Pros:

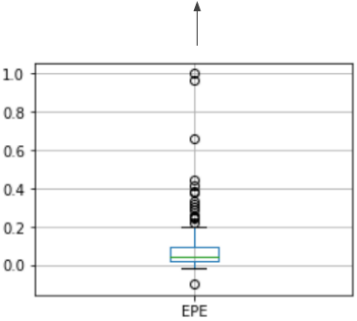
- Higher depth range.
- Fewer missed objects.
- More defined edges.

Cons:

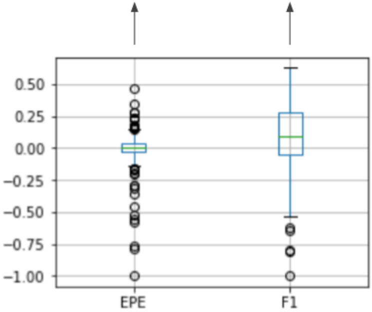
- Overly illuminated stationary surfaces are handled incorrectly. (This is still a remaining problems for all the methods). The reason could be that there's existing bias in the network that overly illuminated objects have the same color as the sky and therefore, they are far.

Flow Results Analysis

The same analysis was done for the Kitti flow 12 and the Kitti Flow 15 datasets.

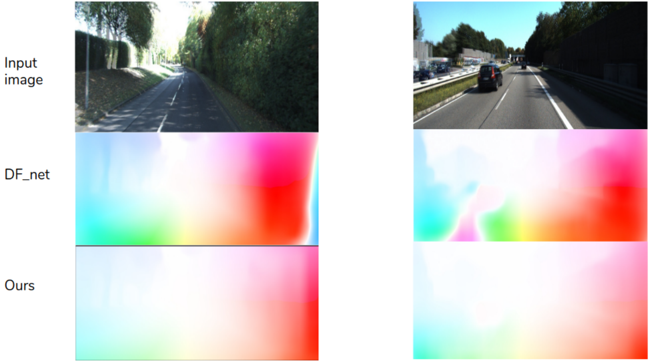


Flow 12 EPE



Flow 15 EPE and F1 score

Flow Results Analysis: Better



Flow Results Analysis: Worse

Flow Results Analysis

Pros:

- Unlike our approach, DF-net produces unexplainable discontinuity in flow field.
- The flow fields produced using this approach are more robust with dynamic objects.

Cons:

- White surfaces produce zero flow. (This is still a remaining problems for all the methods)

Future work

- SSIM could be used instead of ternary loss to measure how far the reconstructed image was similar to the source image.
  - SSIM is more invariant to illumination variations.
- Using second order smoothness loss instead of first order.

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